

Pore-scale imaging of asphaltene-induced pore clogging in carbonate rocks

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ARTICLE INFO

Keywords:

Asphaltene
X-ray microtomography
Pore-scale imaging
Formation damage

ABSTRACT

We propose an experimental methodology to visualize asphaltene precipitation in the pore space of rocks and assess the reduction in permeability. We perform core flooding experiments integrated with X-ray microtomography (micro-CT). The simultaneous injection of pure heptane and crude oil containing asphaltene induces the precipitation of asphaltene in the pore space. The degree of precipitation is controlled by the measurement of differential pressure across the sample. After precipitation, doped heptane is injected to replace the fluid to enhance the contrast between precipitated asphaltene and doped heptane. The micro-CT images are segmented into three phases: void, precipitated asphaltene, and rock. In the experiment, we observed that the precipitated asphaltene which occupied 39.1% of the pore volume caused a 29-fold reduction in permeability. Furthermore, we analyze the spatial distribution of precipitated asphaltene which showed that the asphaltene tended to clog the larger pores. We also computed the flow field numerically on the images and obtained good agreement between simulated and measured permeability. The distribution of local velocity showed that after precipitation the flow was confined to narrow channels in the pore space. This method can be applied to any type of porous system with precipitation.

1. Introduction

The term asphaltene was first defined in 1837 by J.B. Boussingault [1] as the residue of the distillation of bitumen: insoluble in alcohol and soluble in turpentine. Similarly, the present definition of asphaltenes is also based on a solubility class as the n-alkane-insoluble/toluene-soluble fraction of crude oil [2,3].

In oil fields, asphaltenes in crude oil cause problems when they are precipitated. This can occur throughout a production system, i.e., in the reservoir itself, near wellbore, in production tubings, flowlines, and processing facilities. The precipitation of asphaltene can be caused by several factors including changes in pressure, temperature, and crude oil composition. There are mainly two situations which cause asphaltene-related problems in reservoirs and near wellbore flow during the development of oil fields — a decrease in reservoir pressure below the asphaltene onset pressure (AOP), and a change in fluid composition during fluid injection. A decrease in pressure during primary depletion triggers the precipitation of asphaltene, particularly in near wellbore regions where the depletion of pressure is most significant. A change in crude oil composition can also cause asphaltene precipitation in

reservoirs, when crude oil is mixed with injected solvent for improved or enhanced oil recovery such as hydrocarbon gas injection or CO₂ injection. This can happen anywhere in a reservoir where injected solvent contacts crude oil. Reviews of field experience of these asphaltene problems can be found in [4,5].

When asphaltenes are precipitated in the reservoir, they clog the pore spaces of rocks, resulting in a reduction in permeability [6,7]. This permeability reduction has been experimentally observed through the measurement of differential pressure during core flooding on rocks with asphaltene precipitates, for instance [8,9]. The deep bed filtration (DBF) theory is a widely used approach to model the permeability reduction at the Darcy-scale [10]. This model considers the material balance of the volume fraction of asphaltene precipitates, E_A . The net deposition rate for asphaltene is related to the rate of deposition onto the rock surface, entrainment by the rock surface, and plugging of pore throats, with three coefficients corresponding to these mechanisms. E_A determined from the DBF theory is then used to model an associated permeability reduction as a function of reduced porosity. In general, this permeability reduction model is constructed through a match to experimentally observed differential pressure by tuning the coefficients

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<https://doi.org/10.1016/j.fuel.2020.118871>

Received 25 May 2020; Received in revised form 6 July 2020; Accepted 31 July 2020

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in the model [11,12]. However, these numerical modeling approaches which are based on matching pressure signals between experiments and simulations at the Darcy-scale cannot uniquely determine the empirical coefficients in the DBF model. Hence, the observation of asphaltene precipitation and deposition at the pore-scale is important.

In recent years, X-ray microtomography (micro-CT) has been used as a non-destructive technique to visualize organic matter in geological porous systems [13,14]. Feng et al. [13] used micro-CT to visualize precipitated asphaltenes. They used sandstone samples taken from oil fields. The rock samples already had significant amounts of precipitated asphaltenes due to CO₂-steam huff-and-puff performed. Micro-CT imaging was performed on dry samples which contained asphaltene precipitates in the pore spaces. The raw gray-scale images obtained from the dry scan showed a good contrast between precipitated asphaltenes, void space filled with air, and rock matrix. The images segmented into asphaltene, void space, and rock matrix were then used to compute permeability. Although their approach can be used to study the mechanisms of permeability reduction at the pore-scale, differential pressure during the process of asphaltene precipitation and deposition could not be simultaneously obtained, which is important to relate the pore-scale mechanisms to an averaged permeability reduction at the Darcy-scale.

In this study, we extend our capability in imaging of crude oil flowing in the pore space [15] and propose a new experimental procedure which allows the asphaltene clogging in pore spaces to be imaged simultaneously with the measurement of differential pressure. The procedure includes two key developments in this study: a method to have asphaltene precipitation in a controlled manner by monitoring differential pressure along the sample; and an advanced method to visualize asphaltene clogging of the pore space with a micro-CT by introducing a contrast agent. The mixture of crude oil and precipitant is used to clog the pore spaces with precipitated asphaltene, while monitoring the increase in differential pressure. Once the differential pressure reaches a predetermined limit, the fluid in a core sample is replaced with a contrasting fluid which has a higher X-ray attenuation than asphaltene precipitates; then micro-CT imaging is performed. Furthermore, the measured permeabilities before and after asphaltene precipitation are compared with the computed permeabilities on the micro-CT images.

2. Method and materials

2.1. Rock samples and fluid properties

Estailades carbonate was used for the core flooding experiments. This rock is known to have a bimodal pore throat size distribution with two peaks at around 0.6 μm and 10 μm [16]. X-ray diffraction analysis performed on the samples taken from the same block as used in this study showed the mineral composition to be 97.9% calcite and 2.1% quartz (analysis conducted at Weatherford Laboratories, East Grinstead, UK) [17]. The dimension of a core sample used in our flooding experiments was 6.0 mm in diameter and 14.1 mm in length. The permeability of the sample determined from steady-state pure heptane injection was 110.4 ± 3.3 mD, while the resolvable porosity determined from micro-CT imaging performed on the dry sample at a resolution of 2.37 $\mu\text{m}/\text{voxel}$ was 14.7%. The helium porosity measured on a different sample taken from the same block was $30.3\% \pm 0.2\%$, which means that half of the pore volume existed as resolvable macro porosity, while the other half was sub-resolution micro-porosity [17].

The core flooding experiment was performed at a temperature of 40 °C and atmospheric pressure. We used crude oil collected from a stock tank oil in an oil field in Japan where formation damage caused by asphaltene precipitation has been reported [12]. The asphaltene onset pressure (AOP) of the crude oil is estimated to be around 31 MPa at the reservoir temperature. In this field, a small amount of chemical is

Table 1

Summary of the density and viscosity of the fluids used in the study.

	Density	Viscosity
	kg/m ³	mPa·s
Crude oil	830.0 ^{*1}	2.53 ^{*2}
Heptane	666.8 ^{*3}	0.33 ^{*3}
The mixture of crude oil and heptane ^{*4}	748.4 ^{*5}	1.43 ^{*5}
Doped heptane ^{*6}	680.4 ^{*7}	0.48 ^{*7}

^{*1} Measured density at 21 °C and atmospheric pressure conditions.

^{*2} Estimated viscosity for the experimental conditions of 40 °C and atmospheric pressure based on the measured value at 21 °C and atmospheric pressure conditions.

^{*3} Values for the experimental conditions of 40 °C and atmospheric pressure obtained from [19].

^{*4} 50 vol.% and 50 vol.% mixture of crude oil and heptane.

^{*5} Estimated values based on the properties of crude and heptane.

^{*6} 20 wt.% and 80 wt.% mixture of iododecane and heptane.

^{*7} Estimated values based on the properties of dodecane and heptane.

injected into the production lines to prevent asphaltene deposition. Prior to the experiments, the crude oil was vacuum filtered with a Whatman Grade 42 filter with 2.5 μm mesh size to remove any suspended solids in the crude oil. Pure heptane was used as a precipitant. As we describe in detail in Section 2.2, we used three fluids in core flooding: pure heptane, a 50% and 50% mixture in volume of pure heptane and crude oil, and a 80% and 20% mixture in weight of pure heptane and iododecane. The 50–50 mixture was used to cause asphaltene precipitation in the pore space, while the final mixture was used to have a sufficient contrast in X-ray attenuation compared to precipitated asphaltene [18], which will be called doped heptane. The fluid properties at the experimental conditions are shown in Table 1.

2.2. Experimental procedure

Fig. 1 shows a schematic diagram of the experimental apparatus. The experiment was conducted in a Hassler type flow cell made of carbon fiber epoxy with low X-ray photon attenuation. The differential pressure transducer used was a Keller PD-33X, with an accuracy of ± 0.5 kPa [20,21]. The experimental procedure is as follows:

1. Initialization. A confining pressure of 3000 kPa was applied and maintained within the cell to compress the Viton sleeve around the core sample to avoid any fluid bypass. A dry (air) scan was taken at room temperature.
2. Pure heptane injection to measure absolute permeability. The pressure difference was recorded for pure heptane flow at four flow rates ranging from 0.01 ml/min to 0.1 ml/min. A differential pressure before asphaltene precipitation was obtained for a flow rate of 0.02 ml/min. Then the flow was stopped by closing the valves at the inlet and outlet, and micro-CT imaging was performed. This image is termed as the micro-CT image after the first pure heptane flow.
3. Heptane/crude oil mixture injection to induce asphaltene precipitation. Pure heptane and the crude oil were simultaneously injected from each pump at a flow rate of 0.01 ml/min, resulting in the flow of the 50% and 50% mixture by volume of pure heptane and crude oil at a total flow rate of 0.02 ml/min. An increase in differential pressure was recorded during this period. When the differential pressure reached a predetermined pressure limit of 500 kPa, the injection fluid was changed from the mixture to pure heptane.
4. Pure heptane injection to replace crude oil in the core sample. Pure heptane was injected at a flow rate of 0.02 ml/min. After the stabilization of the differential pressure, a differential pressure after asphaltene precipitation was obtained. Then, the flow was stopped

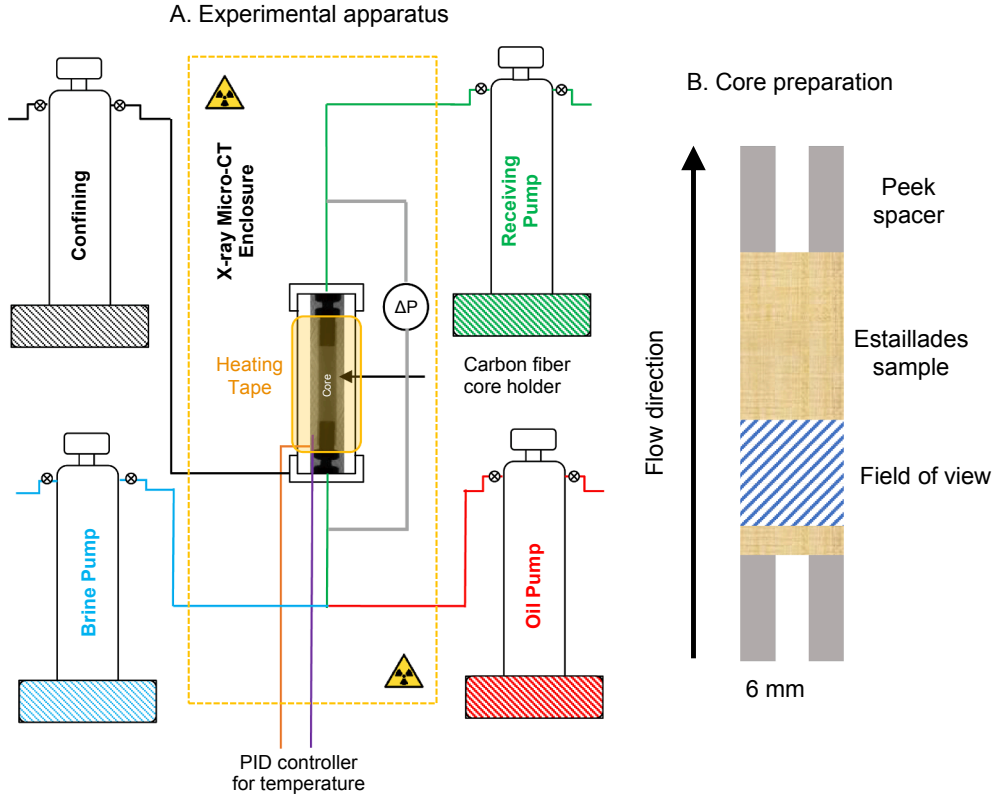


Fig. 1. Schematic diagram of the experimental apparatus. Tomographic scans were taken for the bottom section which was approximately 3 mm above the injection port.

by closing the valves at the inlet and outlet, and micro-CT imaging was performed. This image is called the micro-CT image after the second pure heptane flow.

5. Doped heptane injection to replace pure heptane in the core sample. Doped heptane was injected at a flow rate of 0.02 ml/min. With this step, the pore spaces available to flow were filled with doped heptane, while the precipitated asphaltene remained intact. Then the flow was stopped by closing the valves at the inlet and outlet, and micro-CT imaging was performed. This image is termed the micro-CT image after doped heptane flow.

All the imaging was performed with a Thermo Fisher Heliscan micro-CT. The imaging parameters for each scan were 95 keV (tube current 65 mA), 0.8 s of exposure time, and 3,240 projections. The full size of the imaged cylindrical volume was $2880 \times 2880 \times 2040$ voxels with a voxel size of $2.37 \mu\text{m}$, from which a cubic sub-volume with a size of $1700 \times 1700 \times 1700$ voxels (a $4.0 \times 4.0 \times 4.0 \text{ mm}^3$ cube) was used for image analysis. The imaging location was determined where the distance between the center of the X-ray beam and the bottom of the sample was 5 mm (approximately 3 mm away from the inlet).

In the third step in the above procedure, we saw that crude oil and pure heptane as precipitant first came into contact at a T-junction placed before the inlet. Hence, asphaltene precipitates could appear at the T-junction and flow through a tube with an internal diameter of 0.75 mm. To evaluate an additional pressure drop occurring in the tube, the pressure drop in a tube was measured in separate experiments by flowing the same mixture of the crude oil and heptane in the tube. The observed pressure difference measured for the tube of 0.5 m length was approximately 0.1 kPa, which is two orders of magnitude lower than the pressure difference observed during core flooding, as we will discuss later.

2.3. The Single-Phase Lattice Boltzmann Method

We also evaluated permeability based on direct numerical simulations on micro-CT images. The Navier–Stokes equation for single-phase flow in the pore space was solved using a single-phase lattice Boltzmann (LB) solver constructed for the D3Q19 lattice model, in which the lattice velocity, $\mathbf{e}$, is given as [22,23]:

$$\mathbf{e}_i = \begin{cases} (0, 0, 0), & \text{for } i = 0, \\ (\pm 1, 0, 0)c, (0, \pm 1, 0)c, (0, 0, \pm 1)c, & \text{for } i = 1, \dots, 6, \\ (\pm 1, \pm 1, 0)c, (\pm 1, 0, \pm 1)c, (0, \pm 1, \pm 1)c, & \text{for } i = 7, \dots, 18, \end{cases} \quad (1)$$

where $c = \Delta x / \Delta t$ is the lattice speed with Δx and Δt being the lattice unit length and the time step size, respectively; the subscript i is the index of the lattice velocity taking values from 0 to 18. We set $\Delta x = \Delta t = 1$ in this study.

The particle distribution, f_i , undergoes the following collision processes:

$$f_i^*(\mathbf{x}, t) = f_i(\mathbf{x}, t) + \Omega_i(\mathbf{x}, t) \quad (2)$$

where $f_i^*(\mathbf{x}, t)$ is the post-collision particle distribution; $\Omega_i(\mathbf{x}, t)$ is the collision operator. We used the Multiple-Relaxation-Time collision operator [24] expressed as:

$$\Omega_i = -(\mathbf{M}^{-1}\mathbf{S}\mathbf{M})_{i,j}[f_j - f_j^{eq}], \quad (3)$$

where $\mathbf{M}$ and $\mathbf{S}$ are the transformation matrix and the diagonal relaxation matrix, respectively. The specific form of these matrices can be found in [23]. f_i^{eq} is the equilibrium distribution function which is obtained by a second-order Taylor expansion of the Maxwell–Boltzmann distribution with respect to the local fluid velocity as:

$$f_i^{eq} = \rho \omega_i \left[1 + \frac{\mathbf{e}_i \cdot \mathbf{u}}{c_s^2} + \frac{(\mathbf{e}_i \cdot \mathbf{u})^2}{2c_s^4} - \frac{\mathbf{u}^2}{2c_s^2} \right], \quad (4)$$

where ρ is the fluid density; $\mathbf{u}$ is the local fluid velocity; c_s is the speed of sound and ω_i is the weight coefficient which is given by:

$$\omega_i = \begin{cases} \frac{1}{3}, & \text{for } i = 0, \\ \frac{1}{18}, & \text{for } i = 1, \dots, 6, \\ \frac{1}{36}, & \text{for } i = 7, \dots, 18. \end{cases} \quad (5)$$

Through the Chapman-Enskog expansion analysis of the above scheme, the continuity equation and Navier-Stokes equation are recovered with the following relations:

$$\rho = \sum_i f_i(\mathbf{x}, t), \quad (6)$$

$$\rho \mathbf{u} = \sum_i f_i(\mathbf{x}, t) \mathbf{e}_i, \quad (7)$$

$$\nu = \frac{1}{3} \left(\tau - \frac{1}{2} \right), \quad (8)$$

$$c_s = \frac{1}{\sqrt{3}} c, \quad (9)$$

$$p = \rho (c_s)^2 = \frac{\rho}{3}, \quad (10)$$

where ν is the kinetic viscosity of fluid k ; τ is the relaxation time parameter; p is the hydrodynamic pressure in the fluid.

3. Results

3.1. Differential pressure during core flooding

The observed differential pressure difference during the first pure heptane flow (step 2) where there were no asphaltene precipitates in the system was 0.5 kPa. Fig. 2a shows the measured pressure during the mixture injection (step 3) and the second pure heptane injection (step 4). The pressure increased and reached the predetermined limit of 500 kPa after 2.3 h of the mixture injection. At this point, the injection was changed to pure heptane injection (step 4). However, the differential pressure continued to increase for a while. This is because remaining precipitates and crude oil in the upstream tube continued to flow into the core sample. At 3 h, the differential pressure reached a peak and then it sharply decreased. This corresponds to a breakthrough of pure heptane, indicating that the continuous flow pathways of pure heptane from the inlet to the outlet was established at this point. Fig. 2b shows the observed differential pressure during the second pure heptane injection (step 4) and doped heptane injection (step 5). During the second pure heptane injection where asphaltene precipitated, the observed pressure was 14.5 kPa on average, which is 29 times higher than the pressure difference measured before asphaltene precipitation, indicating a significant reduction in permeability. We see two levels of differential pressure during this period. We consider that a higher differential pressure observed between 0.4 to 1.3 h was caused by the rearrangement of precipitates. During doped heptane injection (step 5), the average pressure was 22 kPa, which is consistent with the viscosity difference between pure and doped heptane, but does not indicate any further change in permeability.

3.2. Image preprocessing

The raw gray scale images are shown in Fig. 3. Although there was a significant difference in the differential pressure between steps 2 and 4, the raw gray scale images corresponding to these steps look similar. This is because the X-ray attenuation of the asphaltene precipitated from the crude oil used in this study was similar to that of the mixture of crude oil and heptane (Figs. 3a and b). After the flooding with doped heptane (step 5), we can see that some pore regions in Fig. 3c were

replaced with doped heptane with a lighter gray, compared to Fig. 3b.

The raw gray scale image obtained after doped heptane flooding was used to quantify the amount of asphaltene precipitated. A non-local means edge preserving filter [25] was applied to the gray scale image to reduce the image noise. Image segmentation into asphaltene precipitates, doped heptane and rock grains was performed using a seeded watershed algorithm, where a seed was generated using a two-dimensional histogram of both the gray scale and the grayscale gradient images [26,27]. The resultant segmented images in which different labels were assigned for the three phases are shown in Fig. 4. The segmented resolvable porosity was 14.7%, among which 39% of the pore volume was occupied with precipitated asphaltene.

Surface coverage of the pore wall was studied based on the segmented images. The boundaries of different phase labels were extracted. Table 2 shows the surface coverage normalized by the total pore surface area with the segmented volume of the phases. 39% pore volume of precipitated asphaltene had a contact area of only 16% of the total pore wall surface area, while 84% of the pore wall surface area was covered with flowing doped heptane. This suggests that precipitated asphaltene existed as solid particles clogging the pore spaces rather than films covering the pore wall surface.

3.3. Pore occupancy of precipitated asphaltene

The pore space was divided into 323,108 individual pore regions based on the distance map, which is the distance between each voxel to the nearest solid. A pore region is defined as a locally wide region in the pore space; its center is the center of the largest sphere that can fit in the pore space. A pore region contains all the void voxels associated with a pore. The algorithm used in this study is similar to that described by other authors – for instance, [28,29]. The disconnected clusters of aggregates were assigned different labels, resulting in 1,014 disconnected clusters. The 3D image of the pore regions and disconnected clusters of asphaltene are shown in Fig. 5.

We now study the pore occupancy of precipitated asphaltene. As shown in Fig. 6a and b, the precipitated asphaltene occupied principally the larger pores. Fig. 6c shows the distribution of all pores and those containing precipitated asphaltene as a function of an equivalent pore radius. The number of pore voxels (doped heptane or asphaltene filled voxels) and asphaltene filled voxels were sampled into equal bins of the equivalent pore radius as shown in the black and red lines in the figure, respectively. Based on these distributions, the fractional occupancy of aggregates for each bin is obtained as shown by the blue line. We see that the clogging by aggregates occurred within relatively larger pore regions with the peak pore size of 100 μm , consistent with the visual observations.

3.4. Permeability change before and after asphaltene precipitation

As discussed in Section 3.1, the pressure difference measured with the flow of pure heptane over the sample length of 14.1 mm was 0.5 kPa and 14.5 kPa for the experimental stages before and after asphaltene precipitation, respectively, corresponding to the measured permeability of 110.4 ± 3.3 mD and 3.8 ± 0.2 mD.

The permeability before asphaltene precipitation was computed on the segmented void spaces. We did not consider the contribution of unresolved porosity, which was below the resolution, to permeability, as it was unavailable from the experiment. For permeability after precipitation, the computation was performed only on the segmented label for pure heptane as shown in blue in Fig. 4. This means that the segmented labels for the rock grains (white in Fig. 4) and precipitated asphaltene (red in Fig. 4) were treated as a solid phase.

The numerical simulations were performed on the entire cubic sub-volume used for the image analysis with a size of $1700 \times 1700 \times 1700$ voxels ($4 \times 4 \times 4$ mm³) which accounted for 1/6 of the volume of the sample used for the flooding experiments. A 100 voxels of void buffer

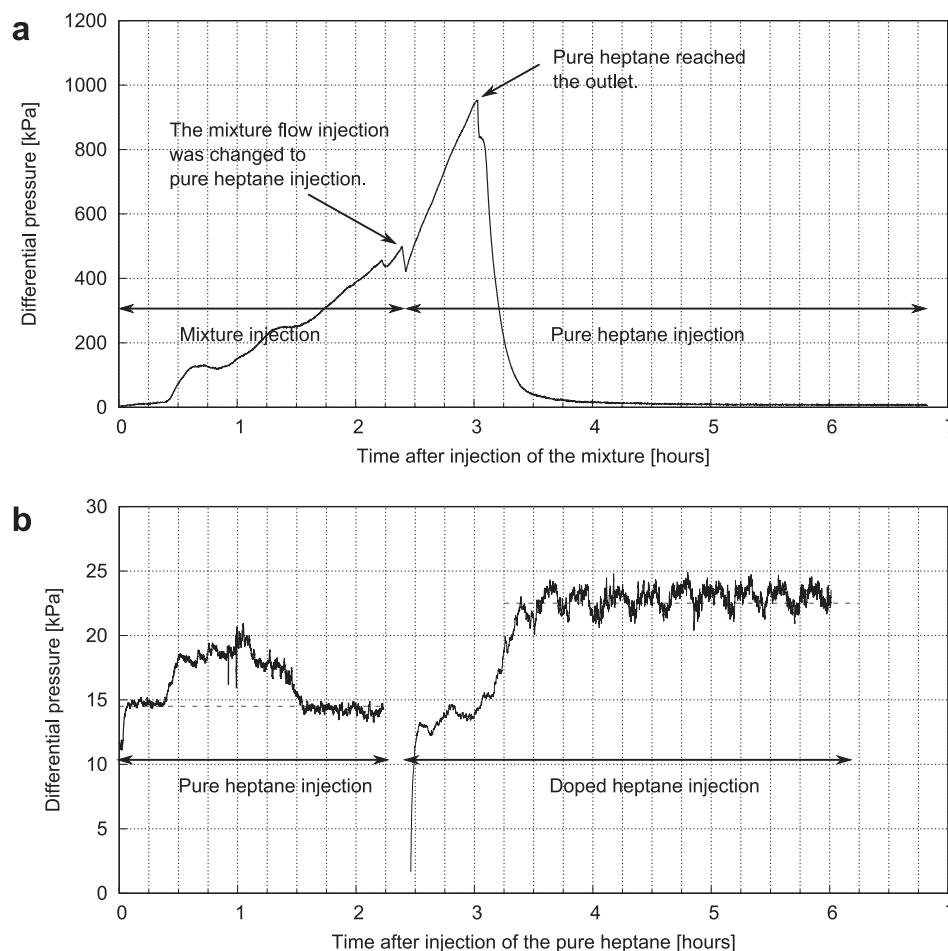


Fig. 2. The measured differential pressure during core flooding.

region was attached to both the inlet and outlet face of the cubic domain in the flow direction to apply a periodic boundary condition. The flow was driven by a constant body force in the flow direction, while a no-slip boundary condition was applied on the solid boundary nodes. The convergence of computed permeability was evaluated at every 1000 time steps, then the simulations were stopped when the relative difference from the last 1000 time steps became smaller than 10^{-3} .

Table 3 compares the measured and computed permeability which

shows reasonable agreement, particularly after precipitation. We consider that the difference in these values was mainly caused by (1) the difference in a domain size for the permeability estimation and (2) the contribution of unresolved porosity. Nevertheless, both of them showed a fairly consistent result — more than a hundred mD permeability before precipitation which decreased to a few mD after precipitation. Figs. 7(a) and (b) show the spatial distribution of the normalized fluid velocity, U_{abs}/U_{avg} , where U_{abs} is the absolute value of computed flow

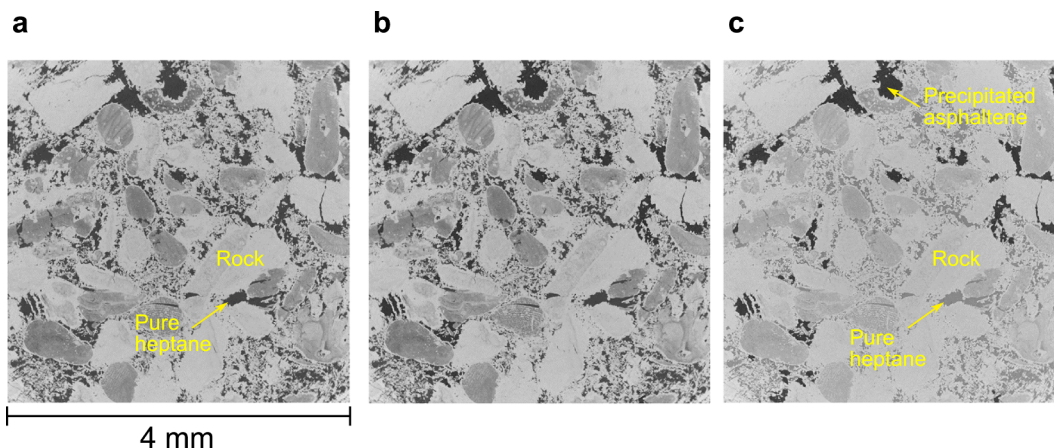


Fig. 3. Micro-CT images during core flooding. Raw gray scale images taken after (a) first pure heptane flooding, (b) second pure heptane flooding, and (c) doped heptane flooding. Here, some pore regions were replaced with lighter gray color, because pure heptane in these pores shown in (b) was replaced with doped heptane with higher X-ray attenuation.

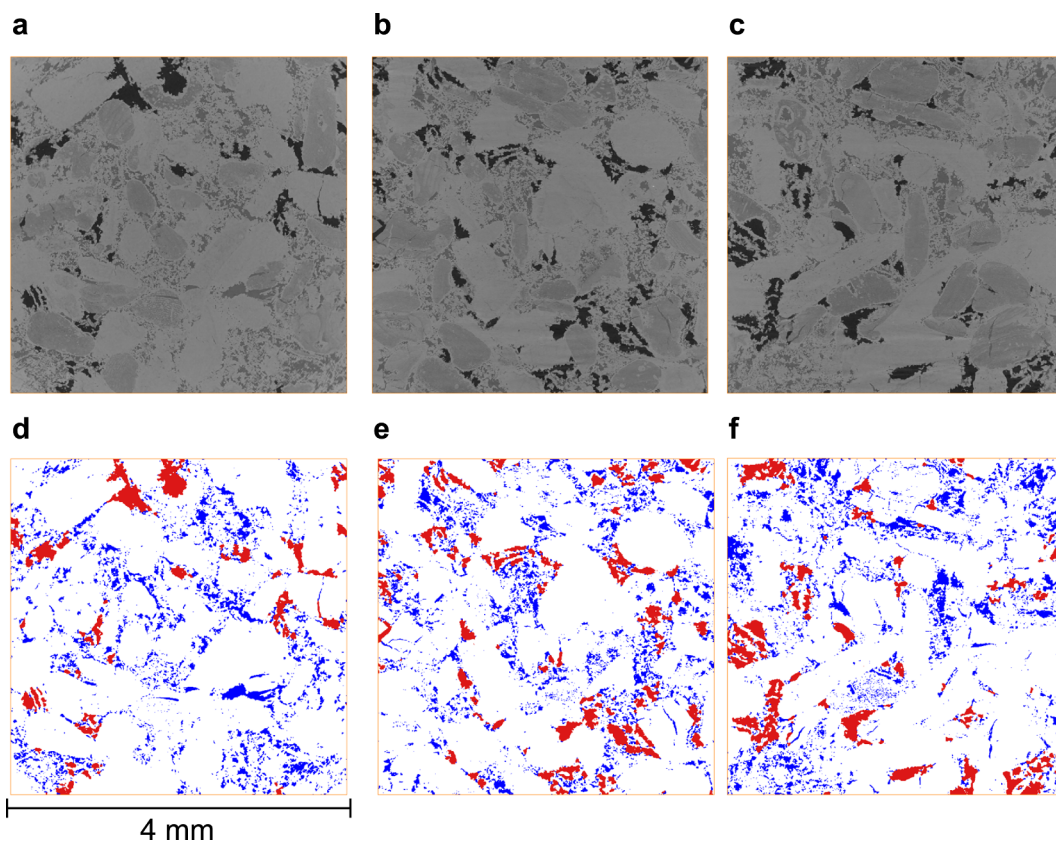


Fig. 4. Segmented images after doped heptane flooding showing (a) an xy slice, (b) a yz slice, and (c) an xz slice of three-dimensional images. The flooding was performed in the z-direction. Here, precipitated asphaltene, flowing heptane, and rock grains are shown in red, blue, and white, respectively.

Table 2

Summary of the volume fraction and pore wall surface coverage of asphaltene and doped heptane. The error bars consider the sensitivity from image segmentation.

	Asphaltene ^{*1}	Doped heptane ^{*2}
Volume fraction	39.1% $\pm$ 1.2%	60.9% $\pm$ 1.8%
Pore wall surface coverage	16.2% $\pm$ 0.9%	83.8% $\pm$ 1.5%

^{*1} Asphaltene clogging the pore spaces present as a solid phase.

^{*2} 20 wt.% and 80 wt.% mixture of iododecane and heptane as a fluid phase.

velocity, and U_{avg} is the average velocity computed on the image before precipitation [30]. As shown in the figures, after asphaltene precipitation, the flow was confined into narrow channels with lower velocity. Fig. 7c shows the histogram of U_{abs}/U_{avg} sampled uniformly in 100 bins. Since asphaltene occupied the center of the pore spaces as shown in Fig. 6c, flow channels with higher velocity observed before precipitation were not seen in the histogram after precipitation. Moreover, there are more lower velocities which are a consequence of the narrow channels formed after asphaltene precipitation.

The visualization of organic matter, such as asphaltene precipitation in the pore space, has been challenging so there are limited studies in

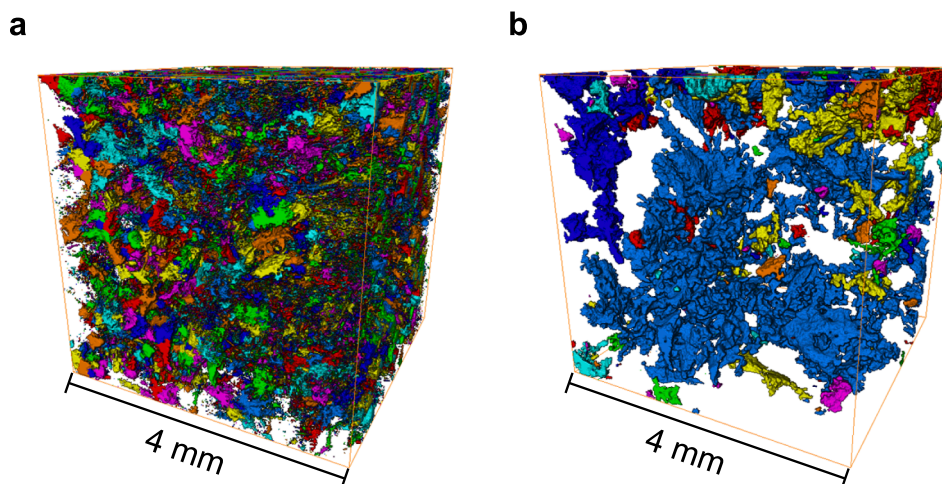


Fig. 5. Identification of (a) individual pore regions and (b) disconnected clusters of precipitated asphaltene indicated by the different colors.

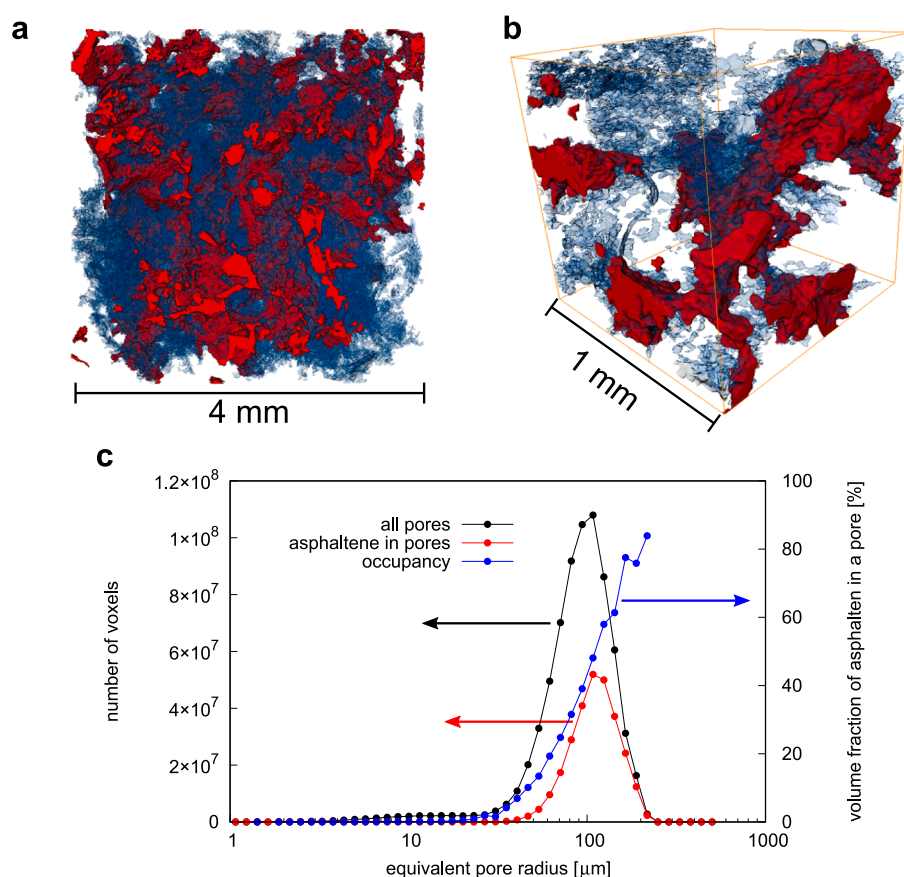


Fig. 6. Pore occupancy of asphaltene that induces clogging. (a) Segmented asphaltene is shown in red within a 4.0 mm cubic volume. The pore wall is shown in transparent blue. The porosity of the volume is 14.7% among which 39% was clogged with asphaltene. (b) An exemplar volume with a size of 1 mm cube taken from the left figure. Clogging by asphaltene occurred in the larger pores, forming connected clusters over multiple pore regions. (c) Distribution of aggregates as a function of the equivalent pore radius. The coloured arrows indicate the relevant values on the y axes. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 3

Comparison of measured and computed permeabilities before and after asphaltene precipitation.

	Experiment ^{*1}	Simulation ^{*2}
	[mD]	[mD]
Before asphaltene precipitation	110.4 ± 3.3	185
After asphaltene precipitation	3.8 ± 0.2	3.45

^{*1} Permeability was evaluated for the entire sample with a size of 6.0 mm in diameter and 14.1 mm in length. The error bars consider uncertainties from the experimental measurements, particularly the sensitivity when measuring differential pressure measurement using a pressure transducer.

^{*2} Permeability was evaluated for the sub-volume used for the image analysis with a size of 4.0 × 4.0 × 4.0 mm³ cube.

the literature. The common approaches included using two-dimension techniques such as scanning and transmission electron microscopy (SEM and TEM) [31,32]; three-dimensional X-ray microtomography [33,13] without other phases in the pore space, or using glass micro-models with a uniformly patterned structure to study asphaltene precipitation where the main concern could be an accurate representation of the pore structure compared to complex geological systems [3]. The main advance of this study lies in development of a new experimental methodology to integrate the flow experiment with three-dimensional high resolution X-ray imaging to visualize asphaltene precipitation in the pore space by introducing contrast agent to differentiate the precipitate from the fluid phases. This integration enables us to visualize and quantify the distribution of asphaltene precipitation and its impact on permeability reduction, and to control degree of precipitation by monitoring the differential pressure simultaneously. Furthermore, by using pore-scale simulation tools, the mechanisms of permeability

reduction can be studied, which is important to provide insights at the pore-scale to an averaged permeability reduction at the Darcy-scale. The methodology in this study could be extended and applied in many different porous systems with precipitation processes.

4. Conclusions

We have demonstrated a new experimental methodology to visualize asphaltene precipitation in the pore spaces of rocks and assess its impact on permeability reduction. The simultaneous injection of precipitant fluid (pure heptane in our study) and crude oil containing asphaltene induced pore clogging by precipitated asphaltene in a rock sample. The degree of clogging by precipitated asphaltene can be controlled by setting different pressure limits at which injection fluid is switched to pure heptane injection. We used the same fluid to measure the pressure difference before and after asphaltene precipitation, which has allowed for an accurate assessment of permeability reduction due to the asphaltene-induced clogging of pore spaces.

The injection of doped heptane after the clogging of pore spaces by precipitated asphaltene has provided a sufficient contrast in X-ray attenuation, which has made it possible to segment the images into three phases, i.e., precipitated asphaltene, doped heptane, and solid matrix. This method can be applied to any type of asphaltene precipitate, even if they have a similar X-ray attenuation to that of oil phase, as seen in the crude oil used in this study.

The differential pressure measured after the precipitation of asphaltene was 29 times higher than that measured before the precipitation, indicating a significant decrease in permeability due to the precipitation of asphaltene. This is supported by the computed permeability on the micro-CT images. The micro-CT images showed that the precipitated asphaltene occupied principally the larger pores. As a

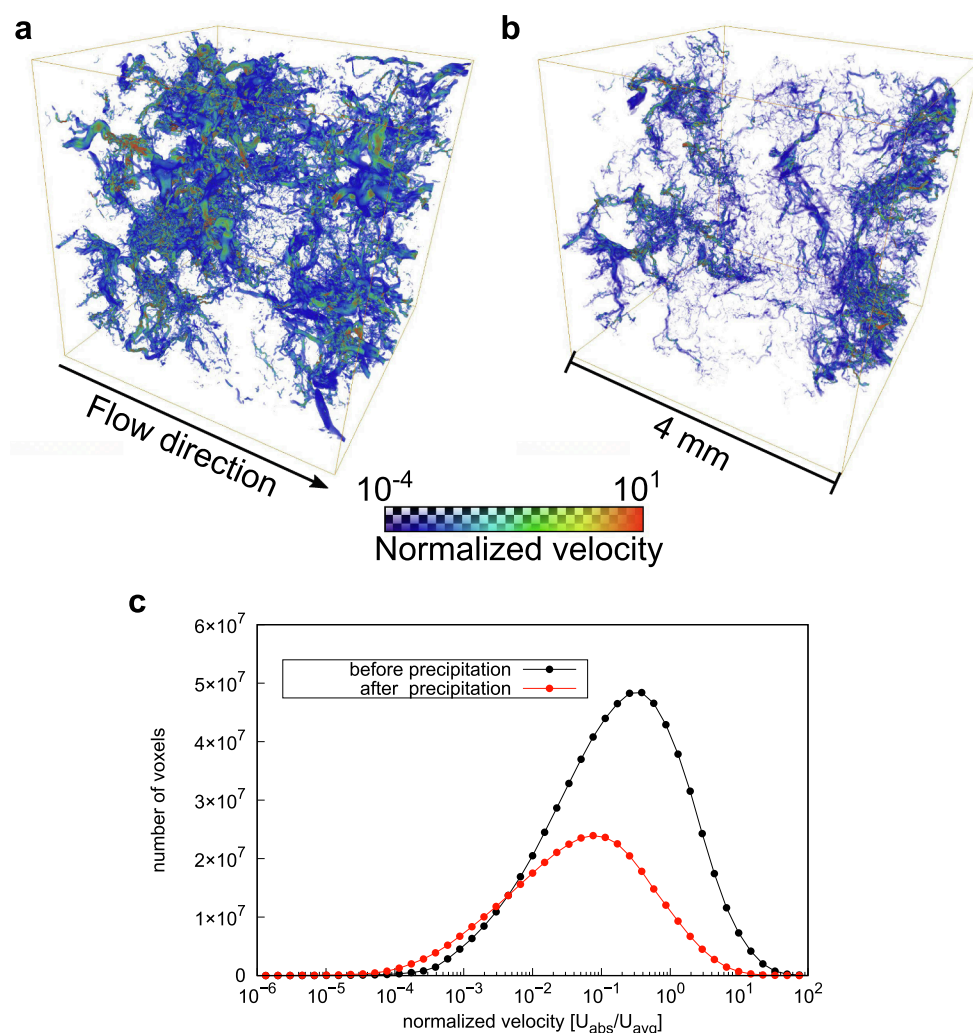


Fig. 7. Computed fluid velocity normalized by the average velocity obtained for the images before precipitation. Distributions are shown for (a) before and (b) after asphaltene precipitation. (c) The histogram of U_{abs}/U_{avg} was sampled uniformly in 100 bins.

result, it was observed in the distribution of computed flow velocity that flow pathways were confined to the narrow channels with low velocity.

In our future work, we will use this experimental methodology to systematically study the mechanisms of permeability reduction due to asphaltene precipitation at the pore-scale. The permeability reduction measured from the experiment will be analyzed as demonstrated in this study.

CRediT authorship contribution statement

Qingyang Lin: Conceptualization, Methodology, Investigation, Data curation, Formal analysis. **Takashi Akai:** Conceptualization, Methodology, Investigation, Data curation, Formal analysis. **Martin J. Blunt:** Conceptualization, Methodology, Investigation, Data curation, Formal analysis, Supervision. **Branko Bijeljic:** Conceptualization, Methodology, Investigation, Data curation, Formal analysis, Supervision. **Hiroki Iwama:** Conceptualization, Methodology, Investigation, Data curation, Formal analysis. **Katsumo Takabayashi:** Conceptualization, Methodology, Investigation, Data curation, Formal analysis. **Yutaka Onaka:** Conceptualization, Methodology, Investigation, Data curation, Formal analysis. **Hideharu Yonebayashi:** Conceptualization, Methodology, Investigation, Data curation, Formal analysis.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

We thank INPEX CORPORATION and Japan Oil, Gas and Metals National Corporation (JOGMEC) for their financial support.

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